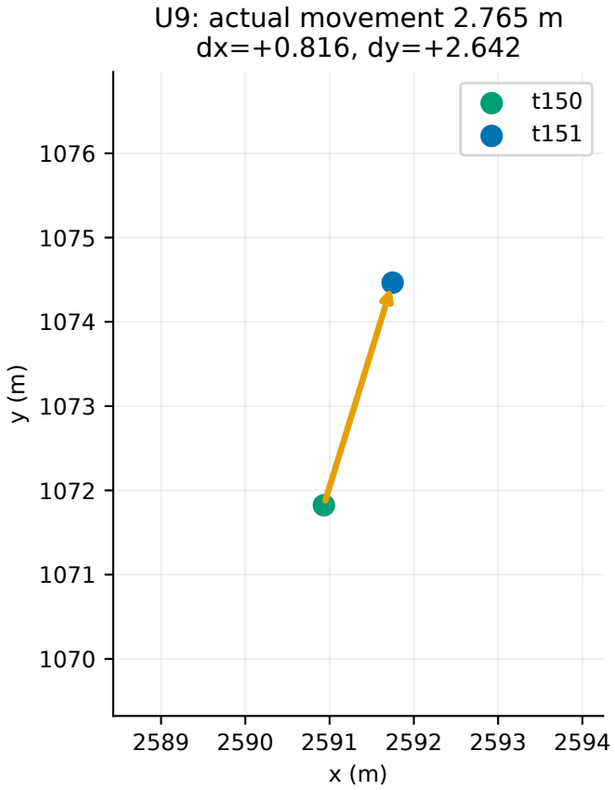
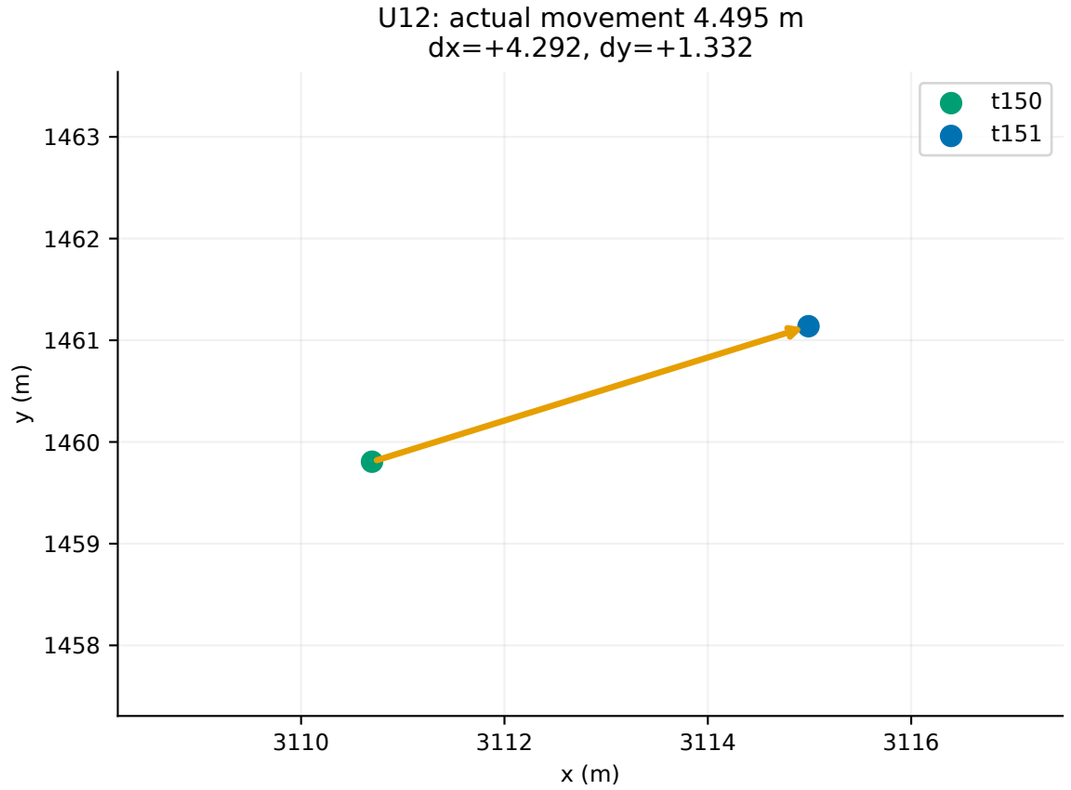
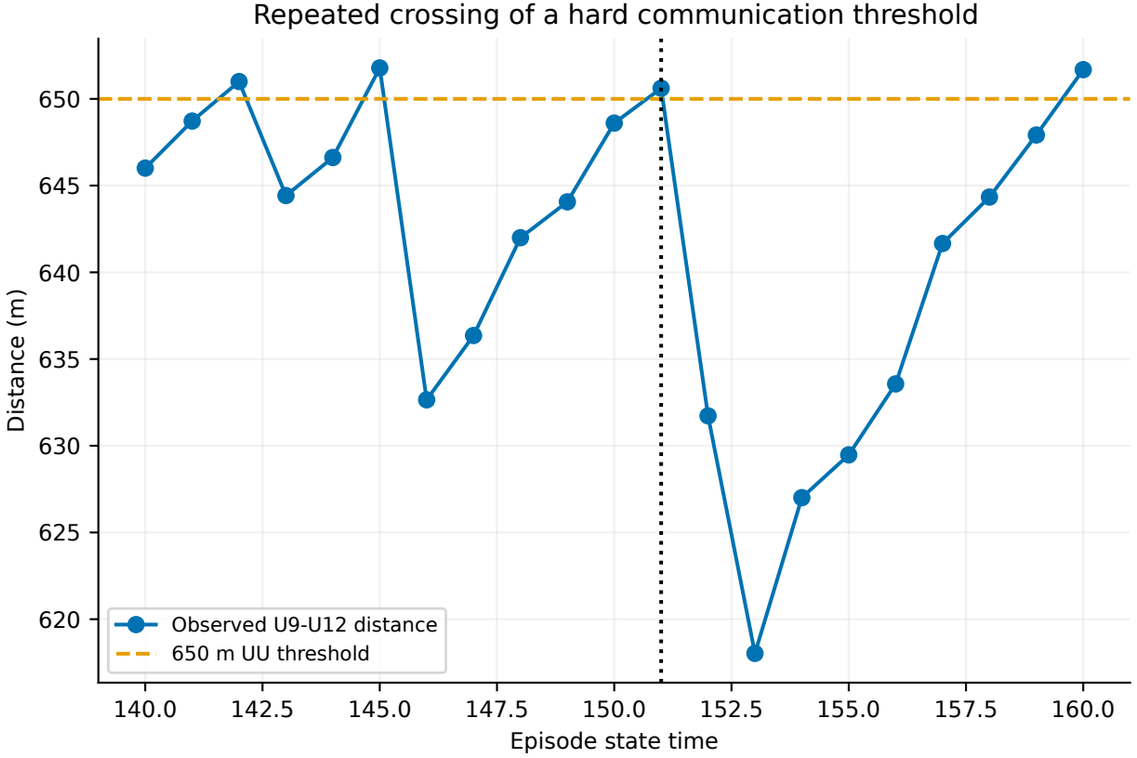
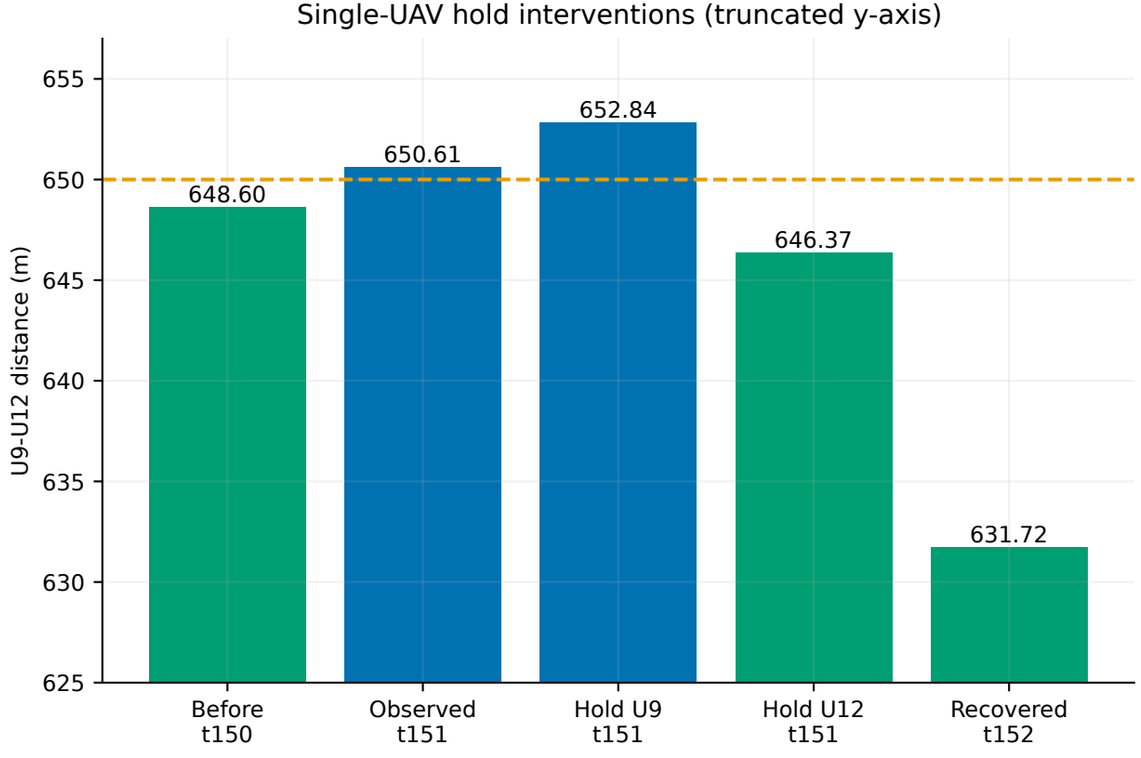

U9: actual movement 2.765 m
dx=+0.816, dy=+2.642
t150
t151
y (m)
x (m)
1076
1075
1074
1073
1072
1071
1070
2589
2590
2591
2592
2593
2594


U12: actual movement 4.495 m
dx=+4.292, dy=+1.332
t150
t151
y (m)
x (m)
1463
1462
1461
1460
1459
1458
3110
3112
3114
3116


Repeated crossing of a hard communication threshold
Observed U9-U12 distance
650 m UU threshold
Distance (m)
Episode state time
650
645
640
635
630
625
620
140.0
142.5
145.0
147.5
150.0
152.5
155.0
157.5
160.0


Single-UAV hold interventions (truncated y-axis)
U9-U12 distance (m)
648.60
650.61
652.84
646.37
631.72
Before
t150
Observed
t151
Hold U9
t151
Hold U12
t151
Recovered
t152
655
650
645
640
635
630
625